# Zhonghe (Catherine) Zheng

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## Education

**Princeton University** Sep 2025 – Present
*Ph.D. Student in Mechanical and Aerospace Engineering*
Advisor: Prof. Anirudha Majumdar

Research: My research focuses on robot learning and world models. I am interested in building systems that can predict and reason about the future to enable more capable and generalizable robot behavior.

**California Institute of Technology** Jun 2025
*B.S. in Computer Science (Machine Learning & AI Track) with a minor in Robotics*

## Research Experience

**IROM Lab** Sep 2025 – Present
*Advisor: Prof. Anirudha Majumdar*

- Creating a controlled study of action-conditioned video world models with an emphasis on controllability, long-horizon prediction, and retention of pretrained visual knowledge
- Developing a novel action-conditioned world model architecture to improve controllability while mitigating catastrophic forgetting when fine-tuning flow models
- Fine-tuned robot policies within a world model trained on autonomously collected play trajectories and conducted real-world hardware validation experiments

**Autonomous Robotics and Control Lab** Oct 2024 – Jun 2025
*PI: Prof. Soon-Jo Chung Mentor: Sorina Lupu*

- Worked on a custom biped platform exploring imitation learning and reinforcement learning for energy-efficient locomotion
- Trained PPO-based locomotion policies in Isaac Lab for energy-efficient bipedal walking

**NASA Jet Propulsion Laboratory — Research Intern** Jun 2024 – Aug 2024
*Mentor: Dr. Ashish Goel*

- Validated a novel hazard avoidance algorithm Optimization Based Kinematic Settling (OBKS) for the Endurance mission
- Developed a scalable evaluation pipeline for validating hazard-avoidance algorithms across multiple simulated terrains
- Integrated OBKS into a state lattice planner and compared performance against the Nav2 planner

**Autonomous Robotics and Control Lab — Research Intern** Jun 2023 – Aug 2023
*PI: Prof. Soon-Jo Chung Mentors: Dr. Matt Anderson, Joshua Cho*

- Led electrical integration for an autonomous drone car prototype
- Integrated emergency power systems and customized flight controller communication pipelines
- Conducted software testing and debugging while improving code modularity
- Experimented with 6-DoF controller for the platform

**CHAOS — NASA Big Idea Challenge** Jan 2023 – Jun 2023

- Contributed to brainstorming, technical development, and proposal writing

## Publications

- Yin, T., Z. Mei, **Z. Zheng**, M. Yamane, D. Wang, J. Sceats, S. M. Bateman, L. Zha, A. Badithela, O. Shorinwa, and A. Majumdar. 2026. "PlayWorld: Learning Robot World Models from Autonomous Play." *Proceedings of the Conference on Robot Learning (CoRL)*, 2026.

- Mei, Z., T. Yin, O. Shorinwa, A. Badithela, **Z. Zheng**, J. Bruno, M. Bland, L. Zha, A. Hancock, J. F. Fisac, and P. Dames. 2026. "Video Generation Models in Robotics: Applications, Research Challenges, Future Directions." *arXiv preprint arXiv:2601.07823*.

- **Zheng, C.** 2024. "Autonomous Flying Ambulance: Hardware Integration and Testing." *Caltech Undergraduate Research Journal*.

- Singh, P. R., N. Ng, A. Patel, C. Garrido, H. Carey, S. Daghlian, R. I. Ray, K. Pulungan, S. Polidoro, **C. Zheng**, and S.-J. Chung. 2023. "SPAMMM – Solar Powered Array for Melting Materials on the Moon." In *2023 Regional Student Conferences*. p. 71914. 2023.

## Teaching Experience

**ME/CS/EE 133A: Robotics — Teaching Assistant** Fall 2024

- Held office hours and recitations covering robotics, linear algebra, and Git
- Explained concepts covering rigid body kinematics, dynamics, Jacobians, and manipulability
- Graded assignments and exams while providing detailed feedback

**ME/CS/EE 129: Experimental Robotics — Teaching Assistant** Spring 2024

- Mentored student teams in robotics software, ROS architectures, and hardware debugging
- Provided weekly technical guidance and code reviews

**CS 21: Decidability and Tractability — Teaching Assistant** Winter 2024

- Assisted students with automata theory, reductions, computability, and NP-completeness
- Held office hours and graded assignments with constructive feedback

## Honors and Awards

**Phillips Second Year Fellowship** 2026
*Princeton University Graduate Fellowship*

**Gordon Wu Fellowship** 2025
*Princeton University Graduate Fellowship*

**Marcella Bonsall SURF Fellowship** 2023
*Summer Undergraduate Research Fellowship*

**Edward C. Stone Voyager SURF Fellowship** 2024
*Summer Undergraduate Research Fellowship @ JPL*

## Presentations

**SURF Seminar Day** Aug 2023

- Presented the Autonomous Flying Ambulance electrical integration project

**Jet Propulsion Laboratory End-of-Project Presentation** Aug 2024

- Presented Endurance project results to the JPL Endurance team

## Skills

**Programming:** Python, C++, ROS2, MuJoCo, Isaac Lab
**Hardware & CAD:** SolidWorks, AutoDesk, 3D Printing, Soldering, Crimping, Laser Cutting

## Leadership & Service

**Health Advocate** Sep 2024 – Jun 2025

- Completed emergency response and CPR training and earned EMR certification
- Served as a health and emergency resource within the student community

**Peer Advocate** Sep 2023 – Jun 2025

- Trained in suicide prevention and peer mental health support
- Supported students experiencing emotional distress and related challenges